

languages, including Python, Java, C++, JavaScript, and Ruby.

OpenAI's GPT is not without contenders in this respect. WuDao 2.0, a multi-modal AI developed by the Beijing Academy of Artificial Intelligence, has around 1.75 trillion parameters and can handle conversations in English and Chinese. The Megatron-Turing Natural Language Generation Model, a collaboration between Microsoft and Nvidia, uses novel parallelism to achieve 530 billion parameters.

BigScience's open-access AI, Bloom, can supposedly handle 46 natural languages and 13 programming languages, with its 176 billion parameters. Google's Language Model for Dialogue Applications (LaMDA), which powers Bard, has 137 billion parameters.

"Yes, there are other powerful transformers and large language models besides GPT-3, but GPT-3 has gained the maximum popularity to date," remarks Azad.

Behind the scenes

Azad spends some time explaining the working of ChatGPT to us. "ChatGPT uses a large neural network to generate responses to user inputs. The neural network is trained on massive amounts of text data and uses this knowledge to generate responses similar in style and content to human-generated text."

He explains the working of ChatGPT in five steps.

Step 1 - Pre-processing: This is the first step to ensure we have the relevant data to feed into the neural network, as neural networks require a lot of computing power to process the information. This includes steps such as tokenisation, where the input is split into individual words, and encoding, where each word is converted into a numerical representation.

Step 2 - Neural network: At its core, a neural network uses a 'transformer architecture' technique to process the input and generate a response. This architecture allows ChatGPT to model complex relationships between words

Demystification zone

Let us first cut through some of the jargon you are likely to encounter in this and other articles on ChatGPT...

Foundation models: We could say that foundation models are, well, foundational to today's AI revolution! Foundation models induce an element of reusability in AI systems, rather than having to create and train task-specific AI models for every single requirement that arises. Foundation models are trained on large, unlabelled datasets and then fine-tuned or trained for specific applications.

"Typically, foundation models are trained using some self-supervised tasks, such as learning, to predict the next word in a sequence of words. Using a foundation model, very little supervised data is required to develop a model for any specific task. A very popular deep learning architecture that is used nowadays for learning language based foundation models is the transformer architecture," says Sachindra Joshi of IBM Research India.

Transformer models: Transformer models track relationships in sequential data and use that to understand context and meaning. Transformers use a technique called self-attention to focus on relevant parts of an input and make more accurate predictions.

"Transformer models are a specific kind of deep learning architecture, which learn the representation of a token (word) based on all the other token representations in a text. This model is very effective in learning good token representations, and the computations required to learn these representations can be easily parallelised using GPUs. This makes it possible to use an enormous amount of textual data to learn very effective token representations," explains Joshi.

(The paid subscriptions to ChatGPT and other AI tools measure your usage based on tokens—now, we have demystified that term as well! According to OpenAI, 1000 tokens would roughly measure up to 750 words.)

Large language models: Large language models (LLMs) are considered to be one of the most successful applications of the transformer model. LLMs are pre-trained using massive volumes of unlabelled data. The LLM learns high-level features at this stage. It can then be fine-tuned for specific tasks.

If you take the example of ChatGPT, it has been pre-trained using troves of textual data from the internet, from which it has learnt to understand context and meaning well enough to hold mostly meaningful conversations with users. As a next step, users can utilise the ChatGPT API to make their own apps, fine-tuned and trained for specific purposes, say, to answer questions on Indian history, the Bhagavad Gita, or customer queries about a company's products.

LLMs can be used for more than natural language processing. They can also be used to study and predict patterns in weather, protein structures, financial data, cyber frauds, and more!

Generative AI: Generative AI is a term that is used to refer to any model that can generate some content—could be text, images, or audio—in response to a prompt. ChatGPT and Dall-E are popular examples in today's context.

Multimodal AI: It can understand and respond in multiple modes, such as text, images, and audio. You could, for example, input an image or a graph as part of your prompt. The latest version from OpenAI, GPT-4, released in March 2023, is multimodal.

and generate grammatically correct and semantically meaningful responses.

Step 3 - Language modelling: The neural network uses a language model to predict the most likely words to follow the user's input. The language model is trained on vast text data and uses this knowledge to make predictions.

Step 4 - Beam search: ChatGPT uses a technique called 'beam search' to ensure that the generated response is grammatically correct and semantically

meaningful. This involves generating multiple possible responses and selecting the one that is most likely to be correct based on a set of criteria, such as the likelihood of each word in the response.

Step 5 - Post-processing: Once the neural network has generated a response, it is post-processed to ensure that it is in a format that is easy for the user to read and understand. This includes steps such as converting